

# Saketh Marrapu

(737)-274-7287 ♦ [saketh.marrapu@gmail.com](mailto:saketh.marrapu@gmail.com) ♦ <https://smarrapu05.github.io/>  
10224 Chestnut Ridge Rd, Austin, TX 78726

## EDUCATION

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| <b>Texas A&amp;M University</b> , College Station, Texas<br><i>Bachelor of Science in Statistics</i><br>Minors in Computer Science, Economics | August 2022 - December 2026<br>GPA: 3.918 |
| <b>Texas A&amp;M University</b> , College Station, Texas<br><i>Master of Science in Data Science</i>                                          | August 2025 - December 2026<br>GPA: 3.918 |

## EXPERIENCE

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| <b>ENDEAVR</b><br><i>Data Science Intern</i> <ul style="list-style-type: none"><li>Worked on the data pipeline for an AI based self-driving model for rural towns</li><li>Created the schema for splitting data into training, testing and evaluation using python</li></ul> | June 2026 – Current |
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## RESEARCH

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| <b>System Dynamics Modeling for Bovine Respiratory Disease</b> <ul style="list-style-type: none"><li>Applied AI modeling for Bovine Respiratory Disease with Adeolu Adekunle</li><li>Built a web dashboard for a vensim system dynamics model with Python and HTML</li></ul>        | June 2026 – Current           |
| <b>BART treeShap Implementation</b> <ul style="list-style-type: none"><li>Expanded understanding in academic implementation under the tutelage of Dr. Huiyan Sang</li><li>Implemented the fast tree SHAP algorithm for Bayesian Additive Regressive Trees using C++ and R</li></ul> | May 2025 – August 2025        |
| <b>SPACES Study</b> <ul style="list-style-type: none"><li>Collaborated with Dr. Tyler Prochnow and a team to implement social network analysis</li><li>Prepared a data transformation pipeline for exploratory data analysis (EDA) using R and Python</li></ul>                     | February 2025 – February 2026 |

## PROJECTS

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| <b>Enzyme Classification Using Graphs</b><br><a href="https://github.com/smarrapu05/stat656final">https://github.com/smarrapu05/stat656final</a> <ul style="list-style-type: none"><li>Explored the performance of graph based traditional machine learning methods to GNNs for classifying enzymes</li><li>Created a GCN and GAT architecture in base pytorch along with the convolution and attention mechanisms</li></ul>      | January 2026 – May 2026     |
| <b>Protein Folding Analysis</b><br><a href="https://github.com/smarrapu05/STAT636final">https://github.com/smarrapu05/STAT636final</a> <ul style="list-style-type: none"><li>Analyzed network statistical properties of proteins and how they fold in R and python</li><li>Implemented adjacency spectral analysis to identify difference in the functions of residues across proteins</li></ul>                                  | August 2025 – December 2025 |
| <b>Analyzing Efficiency of Rail Transport</b><br><a href="https://github.com/smarrapu05/STAT482final">https://github.com/smarrapu05/STAT482final</a> <ul style="list-style-type: none"><li>Researched the relative qualities of rail in the EU across different factors such as safety and travel time</li><li>Extrapolated possible positive effects of implementation of rail in the US with predictive mixed methods</li></ul> | August 2025 – December 2025 |
| <b>Competitive Esport Predictive Analysis</b><br><a href="https://github.com/smarrapu05/valPredictive">https://github.com/smarrapu05/valPredictive</a> <ul style="list-style-type: none"><li>Adapted predictive methods to a new esport (VALORANT) to predict future match outcomes</li><li>Expanded current sports analysis methods to new applications in the esports world</li></ul>                                           | December 2024               |

## ACTIVITIES

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| <b>Index Fund Research</b> <ul style="list-style-type: none"><li>Modeled options trading using Geometric Brownian motion</li><li>Proposed and implemented an index fund motivation based on current market trends</li></ul>                                               | January 2024 – April 2024 |
| <b>CFB Stocks</b> <ul style="list-style-type: none"><li>Helped create an app that tracks college football team performance in the form of stock values</li><li>Implemented a logistic based prediction algorithm to create odds for which team will win a match</li></ul> | January 2025 – May 2025   |

## SKILLS

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| <ul style="list-style-type: none"><li>R</li><li>C++</li><li>Python (PyTorch, Tensorflow, pandas, seaborn)</li><li>SQL</li><li>HTML</li><li>Excel</li></ul> | <ul style="list-style-type: none"><li>Git</li><li>Docker</li><li>AWS</li><li>Linux</li><li>x86-64 Assembly</li></ul> |
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